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Application of machine learning algorithms in an epidemiologic study of mortality

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ABSTRACT

Purpose: Epidemiologic studies are important in assessing risk factors of mortality. Machine learning (ML) is efficient in analyzing multidimensional data to unravel dependencies between risk factors and health outcomes.
Methods: Using a representative sample from the National Health and Nutrition Examination Survey data collected from 2009 to 2016 linked to the National Death Index public-use mortality data through December 31, 2019, we applied logistic, random forests, k-Nearest Neighbors, multivariate adaptive regression splines, support vector machines, extreme gradient boosting, and super learner ML algorithms to study risk factors of all-cause mortality. We evaluated the algorithms using area under the receiver operating curve (AUC-ROC), sensitivity, negative predictive value (NPV) among other metrics and interpreted the results using SHapley Additive exPlanation.
Results: The AUC-ROC ranged from 0.80 — 0.87. The super learner had the highest AUC-ROC of 0.87 (95 % CI, 0.86 — 0.88), sensitivity of 0.86 (95 % CI, 0.84 — 0.88) and NPV of 0.98 (95 % CI, 0.98 — 0.99). Key risk factors of mortality included advanced age, larger waist circumference, male and systolic blood pressure. Being married, high annual household income, and high education level were linked with low risk of mortality.
Conclusions: Machine learning can be used to identify risk factors of mortality, which is critical for individualized targeted interventions in epidemiologic studies.

Introduction

Epidemiologic studies are important in assessing risk factors of health outcomes, such as mortality, in a population of interest. Dietary, demographic, and clinical features are among important risk factors of mortality and often differ between those who are alive (survivors) and those who died (non-survivors). For instance, age is a well-established predictor of mortality, where non-survivors tend to be older than survivors [1,2]. Similarly, Ford and DeStefano [3] showed male sex, current smoking, hypertension, hypercholesterolemia, and physical inactivity to be associated with increased risk for all-cause mortality among participants in an epidemiologic follow-up study. A systematic review and meta-analysis of prospective nutritional studies reported an inverse association of all-cause mortality with consumption of whole grains, fruits and vegetables, nuts, and fish, but a positive association with consumption of red and processed meat[4]. Finding key risk factors that contribute to health outcomes is important in epidemiologic studies that identify individuals at high risk and is critical for devising interventional strategies to mitigate such risks [5]. Currently, traditional models are mostly being used for risk factor assessment. These traditional models, however, are often hampered by nonlinearity and complexities attributable to interactions between risk factors [6]. In the recent years, clinical models developed using artificial intelligence (AI) and machine learning (ML) have gained substantial interest in the healthcare community including in epidemiologic studies [6,7]. Machine learning is a subtype of AI that offers a class of models, that are often more accurate than the conventional regression models [8]. A further advantage of ML is its capability to analyze large data, complex interactions between features, and high dimensional data, and that it can unravel hidden dependencies between risk factors and outcomes [6,9–11]. Recently, AI has been widely applied to explore the early warning predictors of diseases [12,13]. Given the capability of capturing the nonlinear relationships with ML algorithms, many investigators advocate the application of new predictive models based on ML to support appropriate treatment for patients including personalized care rather than traditional approaches [14].

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Currently, there is limited application of these advanced ML algorithms in epidemiologic studies of mortality, especially in identifying individualized risk factors of mortality. Even in cases where ML algorithms have been applied, a further challenge pertains to interpretation and explanation of how features impact the target variable given the black box nature of ML approaches. In this work, we demonstrated the application of commonly used ML algorithms to assess risk factors of mortality. First, we used individual ML algorithms to assess key risk factors of mortality in a nationally representative epidemiologic study. Second, we developed a super learner that combines the fitted ML algorithms and evaluated its performance against the individually fitted ML algorithms in both training and testing datasets. Lastly, we applied SHapley Additive exPlanation (SHAP) to interpret the outputs from the ML algorithms both at aggregate and individualized levels.

Materials and methods

NHANES-NDI mortality linked data

The National Health and Nutrition Examination Survey (NHANES) includes standardized physical examination, laboratory tests, and questionnaires covering health related topics [15]. The NHANES sample is a representative of noninstitutionalized civilian US population, and includes interview on demographic, socioeconomic, dietary, and health-related questions. NHANES includes an interview in the household, followed by an examination in a mobile examination center (MEC). NHANES participants were considered eligible for mortality follow-up if they provided sufficient information at the time of the interview or MEC follow-up. The NHANES data can be linked to the National Death Index (NDI) mortality using respondent's unique sequence number. National Death Index is a centralized database containing death record information in jurisdictional vital records or statistics offices that can be used to identify each person who has died in the United States as well as cause(s) and the manner of death [16]. National Death Index database contains selected mortality data retrieved from death certificates. States and territories in the US provide death certificate data to National Center for Health Statistics of the US Centers for Disease Control and Prevention through the Vital Statistics Cooperative Program[17]. In the NDI database, deaths are classified using the International Classification of Diseases (Ninth Revision (ICD–9) and 10th Revision (ICD–10)) for underlying and multiple causes of death.

We used publicly available NHANES data collected from 2009 to 2016 linked to public-use NDI mortality data through December 31, 2019 for eligible participants aged 18 years and over at the time of the interview [15]. The limited set of variables included in the public-use NDI mortality data were respondent's eligibility status, mortality status, top nine leading underlying causes of death, person-time of follow-up, and unique number for data linkage[15].

Study variables

For this analysis, we considered the following key potential risk factors of mortality. We considered the following main dietary intakes consumed in the past 24 h: fiber intake (gm), sugar intake (gm), energy intake (kcal), protein (gm), total fat (gm), and cholesterol (mg). Two replicates of these dietary intakes data were collected for foods consumed during the 24-hour period prior to the interview to estimate intakes of energy, nutrients, and other food components. The first dietary recall interview was collected in-person in the MEC, and the second interview data were collected by telephone 3–10 days later. For each person, we estimated long-term average intakes for these dietary variables with the mean intakes reported in the two 24HR recalls. We chose these dietary intake variables given their associations with mortality [18–20]. For demographic variables, we considered gender, age at screening (in years), race, marital status, and education level. For anthropometrics variables, we considered body mass index (kg/m^2), arm circumference (cm), and body waist circumference (cm). Overweight and obesity are linked with increased mortality [21,22], and the risk of death usually differ by sex, age, and race/ethnicity [23,24]. The socioeconomic and lifestyle variables considered included annual household income (in US dollars), ratio of family income to poverty and alcohol consumption (gm). Lastly, the clinical variables considered included systolic blood pressure (SBP, mm Hg), diabolic blood pressure (DBP, mm Hg), creatine level (mg/dL), and albumin level (ug/ML). Albumin measurements are used in the diagnosis and treatment of diseases involving the liver and/or kidneys. Increased microalbuminuria is linked with renal disease and may be predictive of nephropathy risk in patients with type 1 and type 2 diabetes and is also associated with hypertension and cardiac disease[25]. Creatinine measurement is useful in the diagnosis and treatment of renal diseases[26]. The outcome was defined as mortality status (assumed alive/deceased) as of December 31, 2019, as shown in the public-use NDI mortality data.

Data preprocessing

For this analysis, we included data for the participants with complete data on the study variables described above. We further divided the data into training subset (NHANES 2009–2014 period, N = 10954, ~76 %) for developing the models and testing subset (NHANES 2015–2016 period, N = 3414, ~24 %) for evaluating the models. In total, 21 predictors and mortality status were considered. The continuous predictor variables were centered and scaled in building the ML models.

Descriptive statistics

We summarized the distribution of predictor variables in the training set and testing set using frequencies and percentages for categorical variables and means and standard deviations for continuous variables. We tested for the differences in the distributions between the two data splits (training and testing datasets) using Pearson Chi-Squared test for the categorical variables, and *t*-test or sign test for the continuous variables. Similar analysis was conducted by comparing the distributions of predictor variables between those who were still alive with those who were deceased by December 31, 2019.

Machine learning algorithms

We applied commonly used ML algorithms to study risk factors of mortality in the literature starting with the logistic regression [27,28]. The commonly used ML algorithms applied in this study included random forest, support vector machine, multivariate adaptive regression splines (MARS), k-Nearest Neighbors, and boosted ML algorithms (adaptive boosting, extreme gradient boosting, and boosted logistic). Random forest involves building a number of decision trees on bootstrapped training samples [29]. In building decision trees, each time a split in a tree is considered, a random sample of a subset of predictors is chosen as split candidates from the full set of predictors. The split is allowed to use only one subset of predictors. Multiple trees are averaged to provide final predictions [1,9]. In fitting the random forest algorithm, we used 5-fold cross validation resampling technique with a parameter tuning of length of 5 to select the number of predictors to be considered for each split (mtry) and selected the corresponding final model with the largest area under the receiver operating curve (AUC-ROC). Support vector machine (SVM) is used for binary classification in a multidimensional space, and it identifies a hyperplane to discriminate between classes [9]. Support vector machine applies a data transformation that project the data into a higher dimensional space to find a separating decision surface [9]. In fitting SVM, we assumed radial basis function kernel, used 5-fold cross validation resampling technique with a parameter tuning length of 5 and selected the final model with the largest AUC-ROC. Multivariate Adaptive Regression Splines (MARS) is a nonparametric method suited to problems of high dimensional data

[30]. It divides the data space into several, possibly overlapping regions and fits truncated splines function in each region [31]. MARS analysis is carried out in three steps. In the first step, predictor variables explaining the outcome variable well are selected stepwise. In the second step, the model is pruned using iteratively a k-fold cross validation and a generalized cross validation (GCV) procedure, resulting in an optimal MARS model. This is done using different sequences of Monte Carlo cross validation. In the last step, the model with the best accuracy is selected as the final output model, which is based on an evaluation of the predictive performance of the different models [31]. In fitting the MARS algorithm, we performed a 5-fold cross validated grid search to identify the optimal combination of hyperparameters, namely, interaction complexity (*degree*, with values ranging from 1 to 3) and the number of terms to retain in the final model (*nprune*, with values ranging from 2 to 10) that maximize the AUC-ROC. The rationale behind k-Nearest Neighbors (kNN) is that when data are projected into a feature space, the class of a given point is most likely the same as its nearest neighbors. kNN performs classification by assigning a point to the class that is most prevalent out of the k points closest to it. The k parameter can either be predetermined or chosen from a given range. If some predictors have much greater ranges than others, often a normalization process will occur before measuring distance to avoid erroneously ignoring variability in small-scale features. Boosting ML algorithms involve growing trees sequentially, where each tree is grown using information from previously grown trees [29]. We considered the following boosted ML algorithms. First, we explored two variants of extreme gradient boosting: (i) that combines a series of regression trees with a small learning rate (xgbTree), and (ii) the variant that further adds dropout techniques from the deep neural net to boosted trees (xgbDART) [32]. Second, we used adaptive boosting that fits a sequence of weak learners on different weighted training data. The adaptive boosting starts by predicting the original data set and gives equal weight to each observation. If prediction is incorrect using the first learner, then it gives higher weight to observations that are predicted incorrectly. Being an iterative process, it continues to add learner(s) until a limit is reached in the number of models or accuracy [33]. Finally, we fitted a boosted logistic regression. Of the four boosted ML algorithms (xgbTree, xgbDART, adaptive boosting and boosted logistic), we selected the final one using AUC-ROC, sensitivity, and specificity. Noteworthy, for each ML algorithm described above, we performed a 5-fold cross validated resampling with a parameter tuning of length of 5 to select the final model parameters with the largest AUC-ROC.

For each model, we selected the top 10 most important variables in predicting mortality in NHANES as measured by the variable importance [34]. For instance, for random forest, the variable importance is calculated using the following steps. First, the prediction accuracy is computed for the out-of-bag sample, and the same is repeated after permuting each feature variable. Second, the difference between the two accuracies is averaged over all trees and normalized by the standard error of the difference. For MARS, variable importance is measured by the reduction in statistic, such as GCV, when each feature variable is added to the model. The measures of variable importance are then scaled to a maximum value of 100, with a higher value implying a more important variable [35]. The final six ML algorithms were combined to

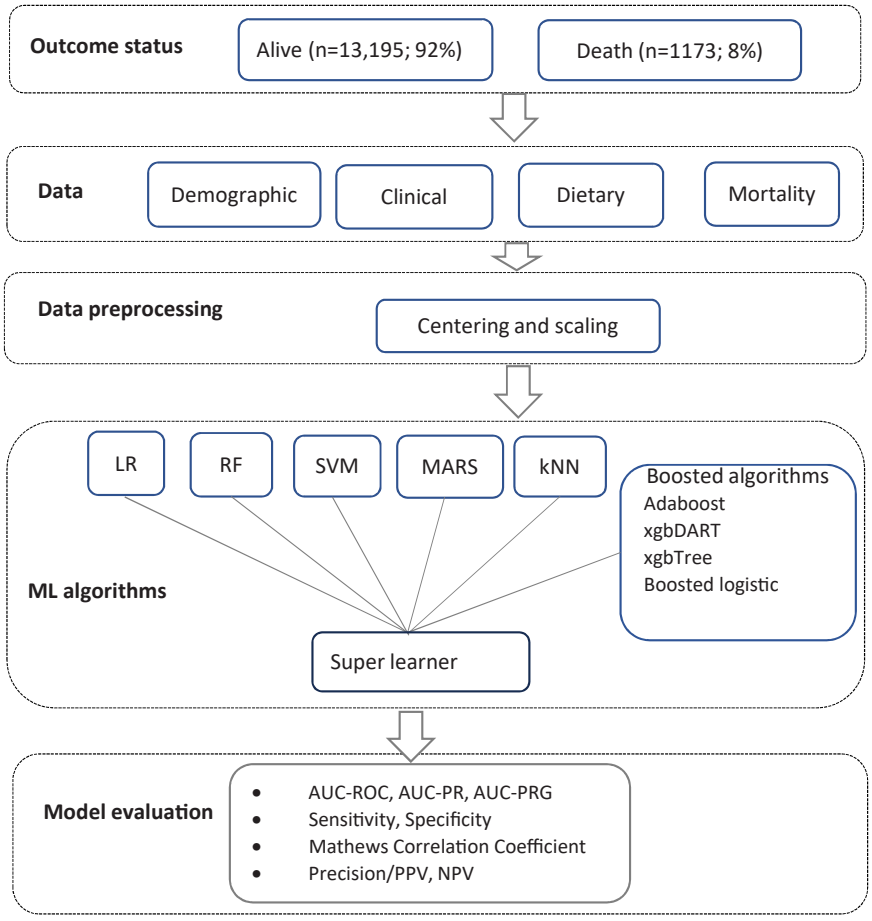


Fig. 1. The flowchart of NHANES-NDI complete data and machine learning algorithms. LR, Logistic Regression; RF, Random Forest; SVM, Support Vector Machine; MARS, Multivariate Adaptive Regression Splines; kNN, k-Nearest Neighbors; xbgDART, extreme gradient boosting with dropout techniques; AUC-ROC, Area Under Receiver Operating Characteristic Curve; AUC-PR, area under precision-recall curve; AUC-PRG, Area Under Precision-Recall Gain; PPV, Positive Predicted Value; NPV, Negative Predictive Value.

Table 1
Distribution of study variables for the NHANES-NDI mortality linked training (2009–2014) and testing (2015–2016) data sets.

Study variables	Training set (2009–2014, N = 10954)	Testing data (2015–2016, N = 3414)	P-value
Sex, n (%)			0.8744
– Female	5136 (46.9)	1606 (47.0)	
– Male	5818 (53.1)	1808 (53.0)	
Race/ethnicity, n (%)			
– Non-Hispanic black	2209 (20.2)	693 (20.3)	
– Another race	8745 (79.8)	2721 (79.7)	
Marital status, n (%)			0.8662
– Married	5595 (51.1)	1736 (50.9)	0.8162
– Not married	5359 (48.9)	1678 (49.2)	
INDHHIN2, mean (Standard Deviation, SD)	8.5 (4.4)	8.8 (4.4)	0.0014
INDFMPIR, mean (SD)	2.6 (1.7)	2.6 (1.6)	0.5222
BMI, mean (SD)	29.0 (6.7)	29.7 (1.6)	< 0.0001
ARMC, mean (SD)	33.3 (5.0)	33.6 (5.2)	0.0031
WAIST, mean (SD)	99.4 (16.4)	101.2 (17.0)	< 0.0001
DBP, mean (SD)	70.7 (12.7)	70.3 (12.7)	0.1077
SBP, mean (SD)	123.6 (18.2)	125.5 (18.3)	< 0.0001
Education level, n (%)			
– Less than 9th grade	769 (7.0)	276 (8.1)	< 0.0001
– 9–11th grade	1540 (14.1)	363 (10.6)	
– High school graduate/GED or equivalent	2427 (22.2)	786 (23.0)	
– Some college or AA degree	3415 (31.2)	1082 (31.7)	
– College graduate or above	2803 (25.6)	907 (26.6)	
Alcohol intake, mean (SD)	10.4 (24.6)	10.0 (22.8)	0.0287
Fiber intake, mean (SD)	17.1 (9.3)	17.0 (9.3)	0.6187
Energy intake, mean (SD)	2115.4 (857.0)	2088.6 (829.4)	0.1024
Protein intake, mean (SD)	83.4 (37.1)	82.1 (36.1)	0.0734
Sugar intake, mean (SD)	112.4 (67.1)	102.9 (60.7)	< 0.0001
Total fat intake, mean (SD)	79.4 (39.2)	82.4 (40.1)	< 0.0001
Cholesterol intake, mean (SD)	294.3 (194.5)	311.0 (206.8)	< 0.0001
Albumin, mean (SD)	36.4 (214.9)	52.4 (343.5)	0.0104
Creatinine, mean (SD)	124.7 (75.0)	126.3 (81.7)	0.2945
Age, mean (SD)	48.3 (17.4)	48.7 (17.2)	0.2384

INDHHIN2, Annual household income; INDFMPIR, Ratio of family income to poverty; BMI, Body mass index; ARMC, arm circumference; DBP, diabolic blood pressure; SBP, systolic blood pressure; n (%), frequency (percent).

obtain a super learner (final ensemble) algorithm [6]. Combining the results of ML algorithms can improve performance of individual ML algorithm by collectively compensating for their limitations, especially when predictions from individual ML algorithms are uncorrelated [6]. We explored linear optimization, random forest, MARS, and xgbDART algorithms to combine the predictions and selected the best combining algorithm using evaluation metrics shown in Figure 1.

Model evaluation: the ML algorithms performance was evaluated in the training set and in the testing set. We evaluated the models using AUC-ROC, area under precision-recall curve (AUC-PR), area under precision-recall gain (AUC-PRG), sensitivity, precision/positive predicted values, Mathew's correlation coefficient (MCC), and negative predicted values [36]. Note that the AUCs are invariant to the cut-off threshold, and that AUC-PR is suited to extreme class imbalance. The

Table 2
Distribution of study variables for the NHANES-NDI linked mortality data by mortality status (live or dead), 2009–2016.

Study variables	Alive (N = 13 195; 92 %)	Dead (N = 1173; 8 %)	P-value
Sex, (n%)			
– Female	6309 (93.6)	433 (6.4)	< 0.0001
– Male	6886 (90.3)	740 (9.7)	
Race/ethnicity, (n%)			
– Non-Hispanic black	2666 (91.9)	236 (8.1)	0.9444
– Another race	10529 (91.8)	937 (8.2)	
Marital status, (n%)			
– Married	6798 (92.7)	533 (7.3)	< 0.0001
– Not married	6397 (90.9)	640 (9.1)	
INDHHIN2, mean (SD)	8.8 (4.4)	6.7 (3.8)	< 0.0001
INDFMPIR, mean (SD)	2.6 (1.7)	2.2 (1.5)	< 0.0001
BMI, mean (SD)	29.2 (6.8)	29.0 (6.7)	0.3887
ARMC, mean (SD)	33.4 (5.1)	32.3 (5.1)	< 0.0001
WAIST, mean (SD)	99.5 (16.5)	103.5 (16.5)	< 0.0001
DBP, mean (SD)	70.9 (12.4)	67.3 (15.1)	< 0.0001
SBP, mean (SD)	123.1 (17.5)	134.3 (23.0)	< 0.0001
Education level, (n%)			
– Less than 9th grade	911 (6.9)	134 (11.4)	< 0.0001
– 9–11th grade	1678 (12.7)	225 (19.2)	
– High school graduate/GED or equivalent	2907 (22.0)	306 (26.1)	
– Some college or AA degree	4183 (31.7)	314 (26.8)	
– College graduate or above	3516 (26.7)	194 (16.5)	
Alcohol, mean (SD)	10.4 (24.0)	9.2 (26.0)	< 0.0001
Fiber, mean (SD)	17.2 (9.4)	15.4 (8.1)	< 0.0001
Energy, mean (SD)	2124.7 (854.8)	1932.7 (780.8)	< 0.0001
Protein, mean (SD)	83.7 (37.1)	75.4 (33.7)	< 0.0001
Sugar, mean (SD)	110.7 (66.1)	103.6 (61.6)	0.0002
Total fat, mean (SD)	80.7 (39.7)	73.8 (36.1)	< 0.0001
Cholesterol, mean (SD)	299.3 (197.9)	286.5 (193.2)	0.0298
Albumin, mean (SD)	32.3 (211.0)	129.4 (515.4)	< 0.0001
Creatinine, mean (SD)	126.4 (77.1)	110.3 (68.5)	< 0.0001
Age, mean (SD)	46.7 (16.6)	67.5 (13.6)	< 0.0001

INDHHIN2, Annual household income; INDFMPIR, Ratio of family income to poverty; BMI, Body mass index; ARMC, arm circumference; DBP, diabolic blood pressure; SBP, systolic blood pressure; n (%), frequency (percent).

other metrics were estimated based on operating point with the maximal informedness due to the class imbalance in the mortality status [36]. Besides the model discriminative measures, we measured calibration by comparing the predicted probabilities with the actual probabilities for the ML models using the calibration curve. The ML algorithms were compared graphically using the AUCs and calibration curves. The schematic representation of data and modeling flow is shown in Figure 1.

We further used Shapley Additive Explanations (SHAP) to explain and interpret the outputs from the fitted ML algorithms both at aggregate and individual level [37,38]. SHAP values provides an objective explanation of how each risk factor impacts the prediction of mortality, where positive SHAP values positively impact the prediction of mortality and vice versa for negative SHAP values. The magnitude of SHAP values quantifies the strength of the effect. We used SHAP force plot for individualized assessment of risk factors and their contributions in predicting all-cause mortality for a specific person. For interpretational comparison of SHAP analysis with the traditional logistic regression, we plotted a forest plot of corresponding odds ratio estimates for the risk factor variables from the logistic regression. The analysis was performed in R software version 4.3.1. The ML models were fitted using CARET package [34] and the model evaluation was done using MLeval [36] package. The SHAP analysis was done using fastshap package [39].

Table 3
Evaluation metrics for the fitted ML algorithms in both training (NHANES 2009-2014) and testing data (NHANES 2015-2016) sets.

	Training dataset						
Metrics	LR	MARS	kNN	RF	xgbDART	SVM	Super learner
AUC-ROC (95 % CI)	0.860 (0.85–0.87)	0.850 (0.84–0.86)	0.810 (0.79–0.83)	0.850 (0.84–0.86)	0.860 (0.85–0.87)	0.800 (0.78–0.82)	**0.870 (0.86–0.88)**
AUC-PR	0.430	0.450	0.360	0.440	**0.460**	0.380	**0.460**
AUC-PRG	0.110	0.110	0.060	0.110	**0.120**	0.080	**0.120**
Sensitivity (95 % CI)	0.792 (0.77–0.82)	0.770 (0.74–0.79)	0.718 (0.69–0.74)	0.808 (0.78–0.83)	0.768 (0.74–0.79)	0.628 (0.60–0.66)	**0.862 (0.84–0.88)**
Specificity (95 % CI)	0.779 (0.77–0.79)	0.788 (0.78–0.80)	0.788 (0.78–0.80)	0.744 (0.74–0.75)	0.804 (0.80–0.81)	**0.867 (0.86–0.87)**	0.717 (0.71–0.73)
MCC	0.375	0.371	0.339	0.351	**0.388**	0.378	0.360
Precision/PPV (95 % CI)	0.274 (0.26–0.29)	0.277 (0.26–0.29)	0.263 (0.25–0.28)	0.250 (0.24–0.26)	0.292 (0.28–0.31)	**0.331 (0.31–0.35)**	0.243 (0.23–0.26)
NPV (95 % CI)	0.973 (0.97–0.98)	0.970 (0.97–0.97)	0.964 (0.96–0.97)	0.974 (0.97–0.98)	0.971 (0.97–0.97)	0.957 (0.95–0.96)	**0.980 (0.98–0.99)**
		Testing dataset					
AUC-ROC (95 % CI)	**0.880 (0.84–0.92)**	**0.880 (0.84–0.92)**	0.830 (0.79–0.87)	0.860 (0.82–0.90)	**0.880 (0.84–0.92)**	0.810 (0.76–0.86)	0.870 (0.83–0.91)
AUC-PR	0.230	0.230	0.170	0.220	0.230	0.200	**0.240**
AUC-PRG	**0.030**	**0.030**	0.010	**0.030**	**0.030**	0.020	**0.030**
Sensitivity (95 % CI)	0.862 (0.79–0.91)	0.792 (0.71–0.85)	**0.900 (0.84–0.94)**	0.823 (0.75–0.88)	0.808 (0.73–0.87)	0.723 (0.64–0.79)	0.846 (0.77–0.90)
Specificity (95 % CI)	0.766 (0.75–0.78)	**0.824 (0.81–0.84)**	0.617 (0.6–0.63)	0.773 (0.76–0.79)	0.818 (0.80–0.83)	0.781 (0.77–0.80)	0.762 (0.75–0.78)
MCC	0.275	0.295	0.202	0.264	**0.296**	0.227	0.265
Precision/PPV (95 % CI)	0.127 (0.11–0.15)	**0.151 (0.13–0.18)**	0.085 (0.07–0.1)	0.126 (0.11–0.15)	0.149 (0.12–0.18)	0.116 (0.10–0.14)	0.123 (0.10–0.15)
NPV (95 % CI)	0.993 (0.99–1)	0.990 (0.99–0.99)	**0.994 (0.99–1)**	0.991 (0.99–0.99)	0.991 (0.99–0.99)	0.986 (0.98–0.99)	0.992 (0.99–1.0)

MARS, Multivariate Adaptive Regression Splines; kNN, k-Nearest Neighbors (kNN); RF, random forest; SVM, support vector machine with radial basis; xbgDART, extreme gradient boosting with dropout technique; LR, logistic regression; AUC-ROC, Area Under Receiver Operating Characteristic Curve; AUC-PR, area under precision-recall curve; AUC-PRG, Area Under Precision-Recall Gain; MCC, Mathew's correlation coefficient; PPV, Positive Predicted Value; NPV, Negative Predictive Value; CI, confidence interval.

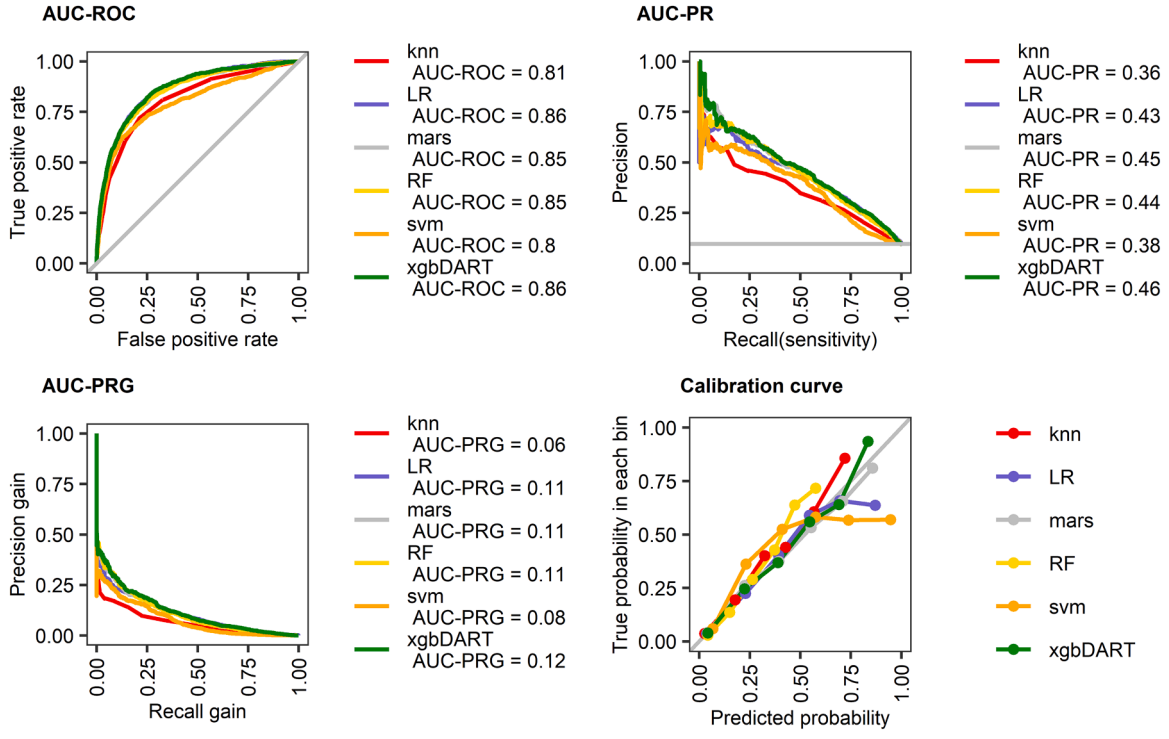


Fig. 2. Discriminative (AUC-ROC, AUC-PR, and AUC-PRG) and calibration (Calibration curve) measures for the five fitted machine learning algorithms to the training dataset: k-Nearest Neighbors (kNN), logistic regression (LR), multivariate Adaptive Regression Splines (MARS), random forest (RF), support vector machine with radial basis (SVM), extreme gradient boosting with dropout technique (xbgDART), NHANES, 2009–2014; ensemble, the super learner; AUC-ROC, Area Under Receiver Operating Characteristic Curve; AUC-PR, area under precision-recall curve; AUC-PRG, Area Under Precision-Recall Gain.

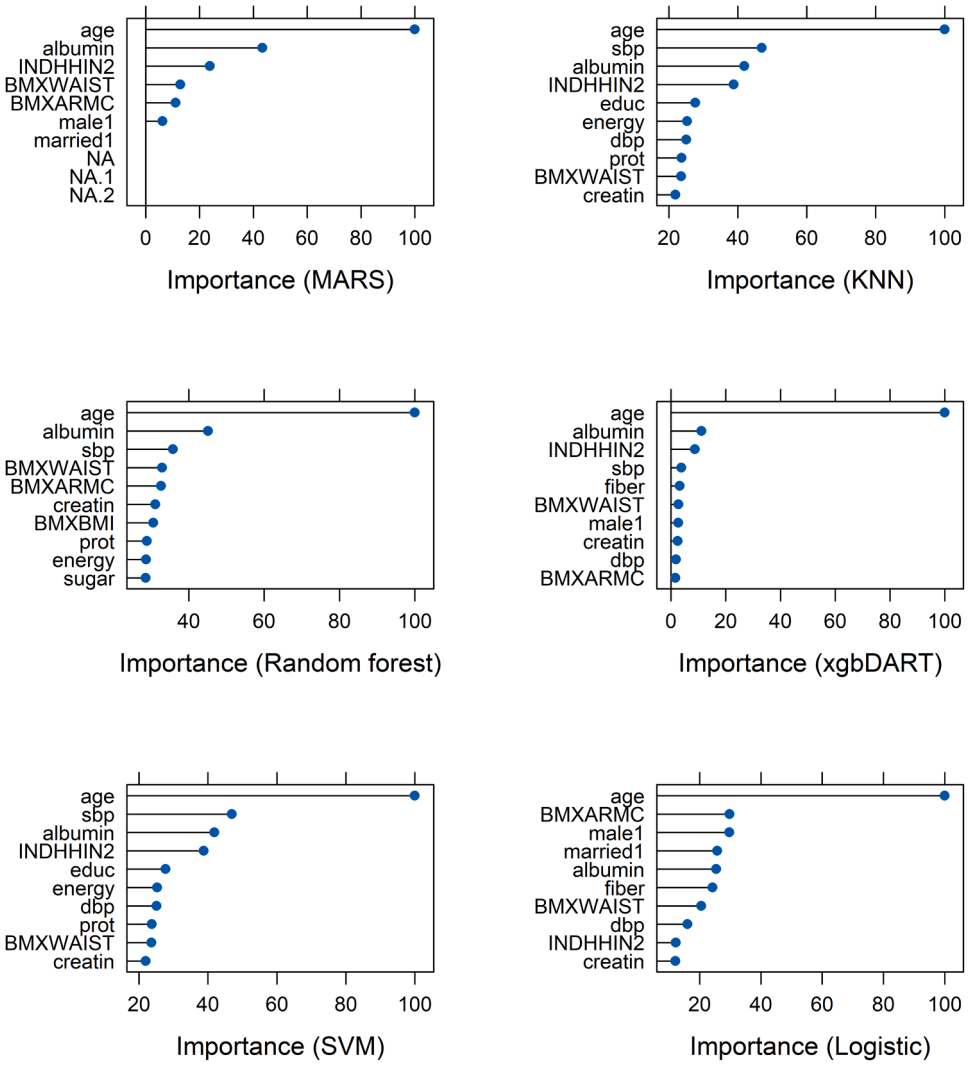


Fig. 3. Top 10 important predictors of all-cause mortality selected by Multivariate Adaptive Regression Splines (MARS), k-Nearest Neighbors (kNN), random forest, extreme gradient boosting with dropout technique (xbgDART), support vector machine with radial basis (SVM), and logistic regression. INDHHIN2, Annual household income; BMXWAIS, waist circumference; BMXARMC, arm circumference; BMXBMI, body mass index; DBP, diabolic blood pressure; SBP, systolic blood pressure; prot, protein intake; NA, no additional features ranked.

Results

Descriptive results

When compared with participants in the training set (NHANES 2009–2014), participants in the testing set (NHANES 2015–2016) had higher mean BMI, arm and waist circumference, household income, fat intake, cholesterol intake, albumin level and education level, but lower alcohol and sugar intake (as shown in Table 1).

Of the 2009–2014 cycle of NHANES participants with complete data on the study variables, 1173 (8 %) had died by December 31, 2019 (as presented in Figure 1, Table 2). We observed seemingly high mortality among males, and unmarried participants. Compared with survivors, deceased participants had high mean systolic blood pressure (SBP), albumin level and age, but lower mean annual household income (INDHHIN2), diabolic blood pressure (DBP), intakes of fiber, energy, protein, sugar, total fat, and cholesterol. We observed similar results when mortality was analyzed within training and testing set; however, there was no statistical difference in the mean distribution of alcohol, dietary intake variables, and creatinine by mortality status in the testing set (see Appendix A).

ML modeling results

The parameter estimates for the fitted ML algorithms were k = 2 for kNN, mtry = 2 for random forest, sigma = 0.0342 and C = 0.5 for support vector machine. For xgbDART, the parameter estimates were nrounds = 100, max_depth = 1, eta = 0.3, gamma = 0, subsample = 0.875, colsample_bytree = 0.6, rate_drop = 0.5, skip_drop = 0.95 and min_child_weight = 1. There was high correlation between outputs from some ML algorithms ($\rho > 0.8$) and were, therefore, standardized in the final ensemble model. The fitted ML algorithms achieved good discrimination. The AUC-ROC ranged from 0.80 — 0.87, AUC-PR from 0.36 — 0.46, and AUC-PRG from 0.06 — 0.12 in the training set. The super learner performed the best in 5 (AUC-ROC of 0.87 (95 % CI, 0.86 — 0.88); AUC-PR of 0.46; AUC-PRG of 0.12; sensitivity of 0.86 (95 % CI, 0.84 — 0.88) and NPV of 0.98 (95 % CI, 0.98 — 0.99)) out of 8 metrics used to evaluate the ML algorithms in the training set. Additionally, the choice of the best performing ML algorithm depended upon the metric of interest. For instance, for the AUC-ROC, sensitivity and NPV, the super learner performed the best in the training set. For AUC-PR and AUC-PRG, xgbDART or the super learner performed the best in the training set (as shown in Table Table 3, Figure 2). Furthermore, when focusing on either specificity or PPV, SVM was preferred. Overall, there was slight

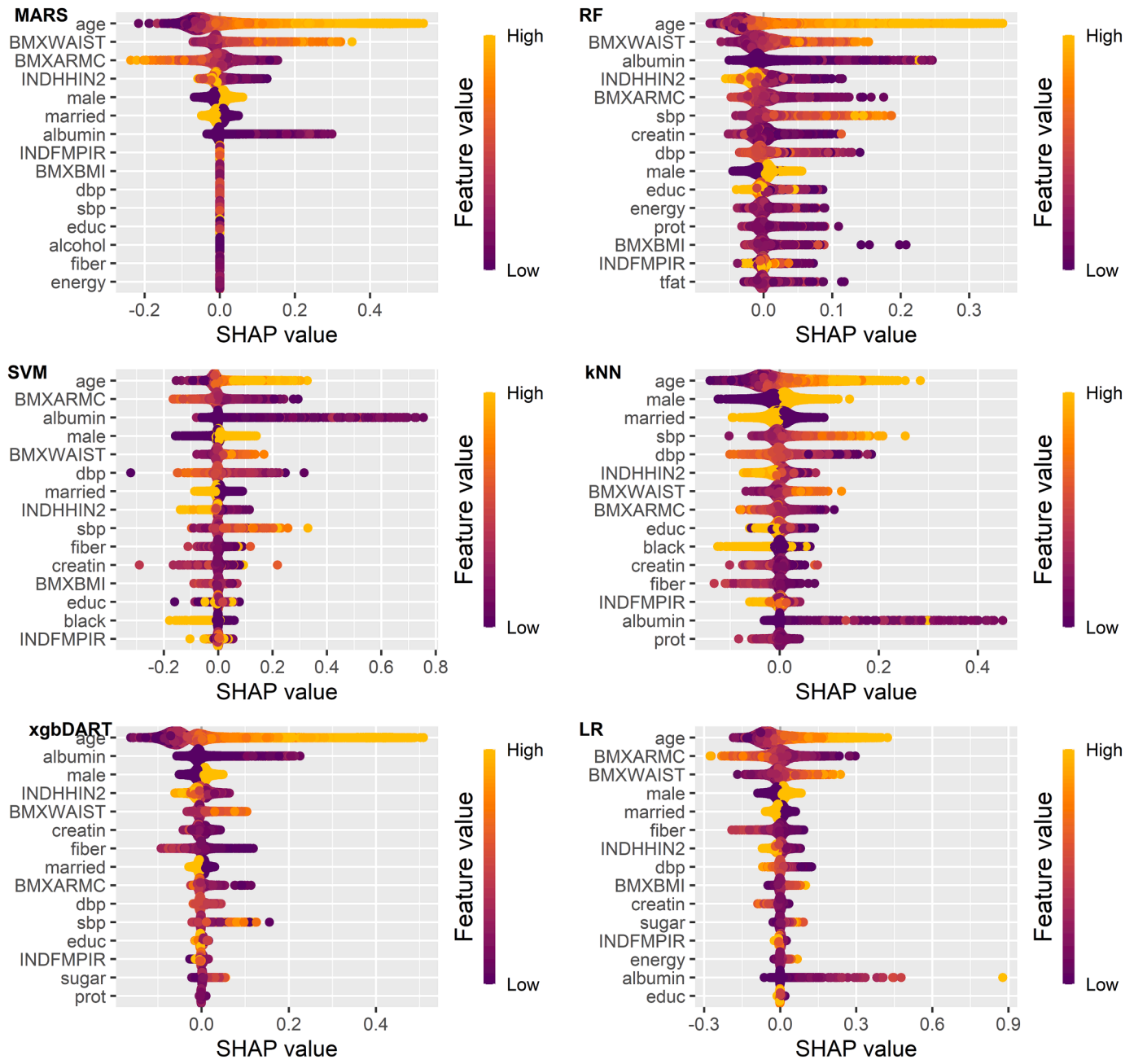


Fig. 4. Shapley Additive Explanations (SHAP) values importance plot to interpret outputs from the 6 ML algorithms. The features are shown in the vertical axis in the order of importance from top to bottom, SHAP values in the horizontal axis indicates change in the log odds and quantifies the impact on the model output, the color of each point represents the value of the corresponding feature with red indicating low value and yellow indicating high value. MARS, Multivariate Adaptive Regression Splines; kNN, k-Nearest Neighbors (kNN); RF, random forest; SVM, support vector machine with radial basis; xbgDART, extreme gradient boosting with dropout technique; LR, logistic regression. INDHHIN2, Annual household income; BMXWAIST, waist circumference; BMXARMC, arm circumference; BMXBMI, body mass index; educ, education level; dbp, diabolic blood pressure; sbp, systolic blood pressure; prot, protein intake; INDFMPIR, Ratio of family income to poverty.

improvement in the performance of the fitted ML algorithms over the simplistic logistic algorithm in both training and testing sets. The ML algorithms were well calibrated in the training set, except for SVM and LR at higher probabilities ≥ 0.75 as shown in the calibration curve that plots true versus predicted probabilities of all-cause mortality (Figure 2). Regardless of the ML algorithm used, age at screening ranked the most important predictor of mortality (Figure 3, Figure 4). However, the other predictors ranked differently depending upon the ML algorithm used. From the SHAP analyses, advanced age at screening, larger waist circumference, being male and SBP were mainly associated with higher risk of all-cause mortality (Figure 4). In contrast, being married, having high annual household income, and high level of education were mainly linked with lower risk of all-cause mortality. The magnitude and direction of effects for most predictors of mortality from the SHAP analysis for logistic algorithm are consistent with those from the traditional logistic regression (Figure 5). For instance, using the traditional logistic regression, age at screening is shown to be highly positively associated (OR= 3.86) with all-cause mortality and vice versa for arm circumference (OR = 0.56). However, traditional logistic regression ranked male gender above waist circumference as top three risk factors of mortality contrary to the findings from the SHAP analysis. Figure 6 presents force plots for personalized risk factors assessment for a randomly selected person who had died by December 31, 2019, in the training set. The f(x) value is the ML algorithm's mortality score for the person and E[f(x)] is

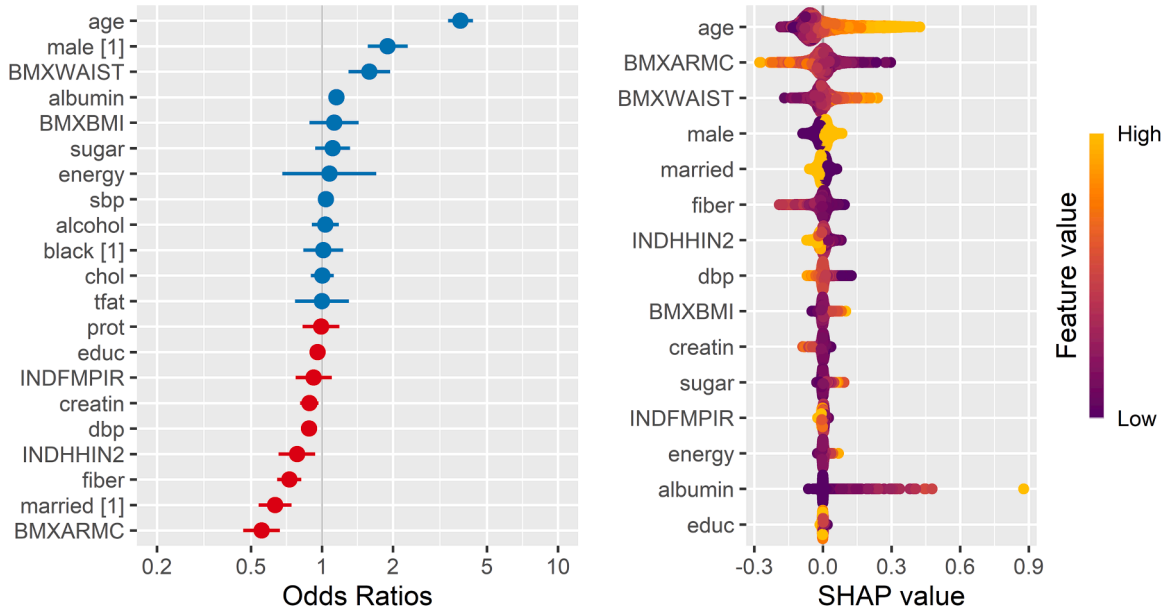


Fig. 5. Forest plot and SHAP plot from the logistic model. INDHHIN2, Annual household income; BMXWAIST, waist circumference; BMXARMC, arm circumference; BMXBMI, body mass index; educ, education level; dbp, diabolic blood pressure; sbp, systolic blood pressure; prot, protein intake; INDFMPIR, Ratio of family income to poverty.

the average mortality score for all persons in the training set. Ideally, a deceased person's f(x) should be greater than E[f(x)]. A higher score leads to the prediction of all-cause mortality. The features that were important predictors of mortality for this person are shown in yellow and purple, with yellow representing risk factors that pushed the mortality score higher, and vice versa for purple. Risk factors with more impact on the mortality score are located closer to the dividing boundary between yellow and purple, and the size of the impact is represented by the size of the bar. The impact of the risk factors on all-cause mortality for an individual varies by the ML algorithm used. From these force plots, advanced age at screening (80 years) had the most impact on mortality score for this person regardless of the ML algorithm used. Other risk factors that contributed to his death varied by the algorithm used, but mainly included his low annual household income (INDHHIN2 =4), being male, having large waist circumference (114 cm), and being unmarried. His mortality SHAP score was greater than the average score of all individuals in the study.

Discussion

We applied commonly used ML algorithms (logistic, random forest, multivariate adaptive regression splines, support vector machine and extreme gradient boosting with drop out technique) with varied degree of complexities to discriminate between NHANES 2009–2016 participants who had died and those who were still alive by 2019. Further, we developed a super learner that combined outputs from these independent ML algorithms with the hope of improving the performance over the individual ML classifiers. With these ML algorithms, we identified key risk factors of mortality among NHANES study participants and provided interpretation and explanation of how these risk factors impact the risk of mortality both at aggregate and individual levels using SHAP.

The main findings from this work were four-fold. First, we found that a super learner performed the best in terms of AUC-ROC, AUC-PR, AUC-PRG, sensitivity and NPV over the individual ML algorithms out of the 8-evaluation metrics used in the training set, but with no discernible improvement in the testing set. Additionally, the complexity of individual ML algorithm did not necessarily translate into improved performance in predicting all-cause mortality for NHANES participants based on the considered evaluation metrics, and that a simpler algorithm may be preferred for ease of interpretability and computational advantage. Second, we learned that the performance of the chosen ML algorithms might be sensitive to the evaluation metric under consideration and the analytic dataset, and that the choice of the ML algorithm may depend on the research question. For instance, the super learner performed optimally in terms of the three AUCs and sensitivity but dismally in terms of PPV in the training set. Third, advanced age at screening was identified as the most important predictor of all-cause mortality among NHANES participants, and that the other predictors of mortality ranked differently depending on the person and the ML algorithm under consideration. Lastly, from the SHAP analysis, we found that advanced age at screening, larger waist circumference, being male and SBP were linked with higher risk of all-cause mortality, whereas being married, having high annual household income, and high level of education were linked with lower risk of all-cause mortality.

Machine learning algorithms are suited for multidimensional data [6]. We, however, considered moderate number of features (n = 21); therefore, sophisticated ML algorithms considered here might not improve the predictive power, and that could explain the observed minimal improvement in performance over the simplistic logistic algorithm. Simpler ML algorithms, such as logistic regression, have been shown to perform similarly to or better than more advanced methods, depending on the data set [40]. Noteworthy, for imbalanced data, where the interest is in the positive class, such as mortality in NHANES, the ROC might give an overly optimistic result. Therefore, precision-recall curve or the precision-recall gain curve, and the area under these may be preferred metrics [36]. Moreover, for highly imbalanced data, using a cut-off probability threshold of 0.5 to get evaluation metrics may not be plausible, prompting the use of operating point with maximal informedness, as is the case in this work, or other metrics such as MCC.

Our findings on the risk factors of all-cause mortality are consistent with previous findings [15,41,42]. For instance, in Mirel, El Bural Felix [15], NHANES participants who were older, had less than high school level of education, never-married, and male gender had a higher risk of mortality than their counterparts. Moreover, a pooled analysis of 11 cohort studies with over half a million participants together with other epidemiologic studies showed a strong association of waist circumference [41,43,44] and SBP [42,45] with all-cause mortality. A comprehensive explanation on how education and income reduce the risk of

mortality is well documented [46]. However, the SHAP analysis from SVM and kNN showed non-Hispanic black participants to have lower risk of death than other ethnicities, which contrasts previous findings from NHANES [15]. This accentuates the level of cautiousness required while interpreting results from ML algorithms, and the risk of obtaining invalid inference from using only one ML algorithm.

The ML-based approaches utilized here and, in the literature, offer effective way of predicting individualized mortality/disease risk, which has significant public health and clinical implications [11]. Noteworthy, ascertaining the key risk factors that contribute to an individual's mortality, such as advanced age and SBP as shown in this study, may benefit public health and/or clinical practitioners in identifying individuals at high risk to target specific intervention or treatment, potentially reducing fatality rate [1], and may help in drawing epidemiological hypotheses [38]. Moreover, the applied interpretable ML approaches provide insights into the complex, non-linear relationships between mortality and its risk factors that are difficult to uncover with the traditional statistical models. Importantly, our ML approaches for individualized risk prediction may help individuals improve self-awareness of their health status. Epidemiologic studies may benefit from ML approaches used here and, in the literature, by leveraging their capability of analyzing complex datasets, and in risk stratification through identification of individuals at higher risk of mortality. Additionally, ML techniques may improve identification of causal relationships in observational studies, with substantial ability to reduce bias and control for confounding, especially in high dimensional data [47]. For instance, in using propensity scores method to estimate causal effects in observational studies, ML algorithms are shown to perform well in the presence of high dimensional data and model misspecification [48]. Future epidemiologic studies may leverage ML techniques in identification of key risk factors of health outcomes using approaches, such as feature importance, as demonstrated in this work. Importantly, ML techniques may improve epidemiologic study designs by enabling informed data-driven decision making, leading to enhanced impact and reliability of research findings.

This study had a few limitations. First, although we included a few clinical variables, information on other relevant risk factors of mortality, such as genetic/neurological markers and medical history, were not collected in NHANES and not included in the analyses. Machine learning model performance may benefit from the inclusion of these important

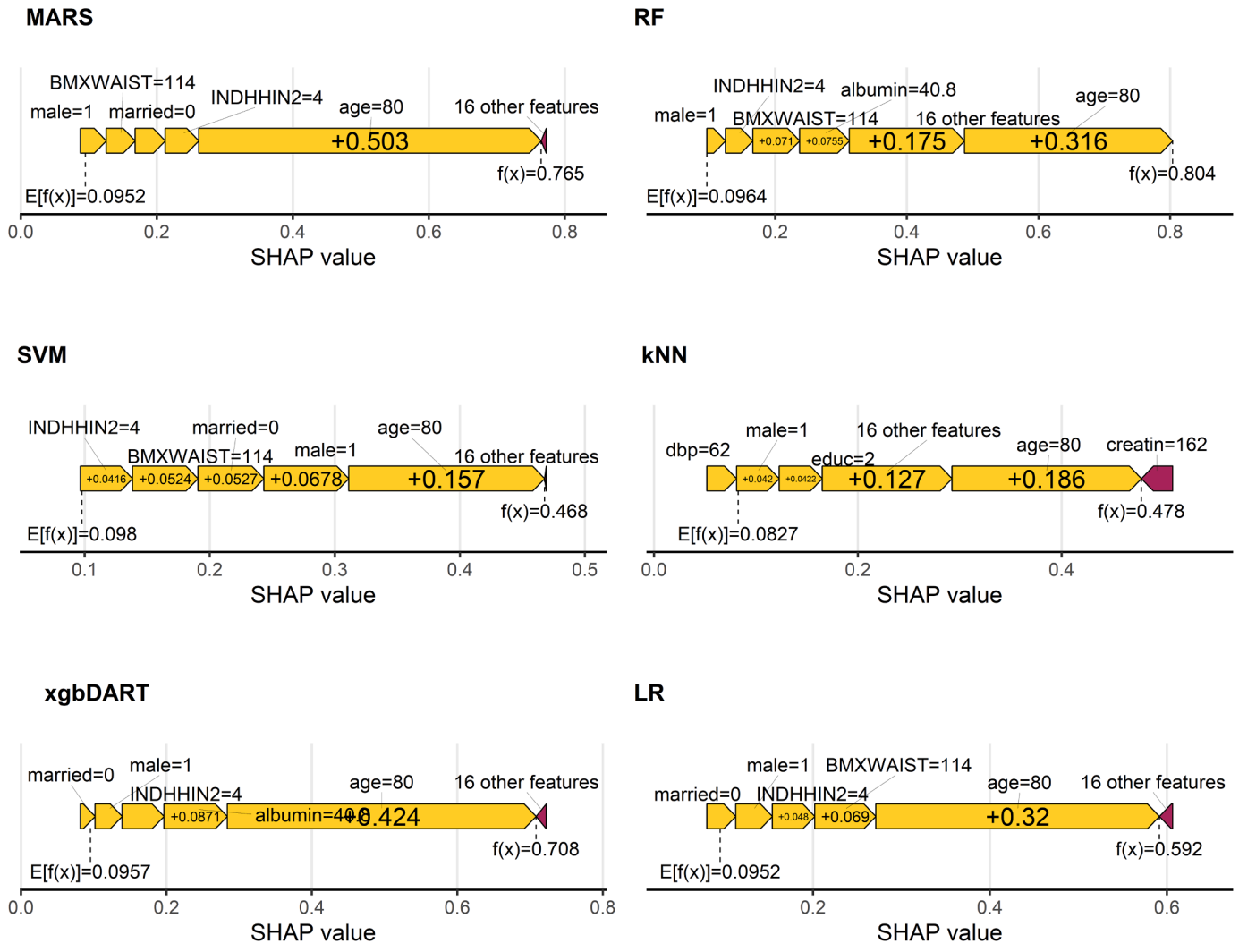


Fig. 6. SHAP force plots for a randomly selected person who was deceased by December 2019 from the fitted ML algorithms. The f(x) value is the ML algorithm's mortality score for the person and E[f(x)] is the average mortality score for all persons in the training set. Important predictors of mortality are shown in yellow and purple, with yellow representing features that pushed the mortality score higher, and vice versa for purple. MARS, Multivariate Adaptive Regression Splines; kNN, k-Nearest Neighbors (kNN); RF, random forest; SVM, support vector machine with radial basis; xbgDART, extreme gradient boosting with dropout technique; LR, logistic regression. INDHHIN2, Annual household income; BMXWAIST, waist circumference; BMXARMC, arm circumference; BMXBMI, body mass index; educ, education level; dbp, diabolic blood pressure.

risk factors. Second, the study was not externally validated in an independent dataset, and therefore, the findings may not be generalizable to non-NHANES participants, especially outside the US. Lastly, the analysis was based only on participants with complete information on the study variables that might further impacts generalizability. For instance, there was higher percent mortality among respondents with observed study variables than those with missing study variables for most risk factors of mortality (see Appendix B). The missing data can be handled with various analytic methods, such as imputation, currently available in the literature [34].

In conclusion, ML algorithms provide efficient approach for aggregated and individualized risk factors assessment of health outcomes in epidemiologic studies. The SHAP analyses offer objective interpretation and explanation of ML outputs, further enhancing their utility in epidemiologic research.

CRediT authorship contribution statement

George Agogo: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Henry Mwambi:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of Competing Interest

All the authors of the manuscript concur with this submission. This manuscript has not been submitted elsewhere for a publication and does not contain experiments with human studies.

All the authors have no conflict of interest to declare for this research.

Distribution of study variables for the NHANES (2009–2016)-NDI linked mortality (through December 31, 2019) data by mortality status in the training and testing datasets

	Train dataset			Testing dataset		
Study variables	Alive	Dead	P-value	Alive	Dead	P-value
Sex, (n%)						
- Female	4743 (92.4)	393 (7.7)	< 0.0001	1566 (97.5)	40 (2.5)	0.0002
- Male	5168 (88.8)	650 (11.2)		1718 (95.0)	90 (5.0)	
Race/ethnicity, (n%)						
- Non-Hispanic black	2001 (90.6)	208 (9.4)	0.8499	665 (96.0)	28 (4.0)	0.72101
- Another race	7910 (90.5)	835 (9.6)		2619 (96.3)	102 (3.8)	
Marital status, (n%)						
- Married	5118 (91.5)	477 (8.5)	0.0003	1680 (96.8)	56 (3.2)	0.0707
- Not married	4793 (89.4)	566 (10.6)		1604 (95.6)	74 (4.4)	
INDHHIN2, mean (SD)	8.7 (4.5)	6.7 (3.9)	< 0.0001	8.9 (4.4)	6.1 (3.5)	< 0.0001
INDFMPIR, mean (SD)	2.6 (1.7)	2.2 (1.5)	< 0.0001	2.6 (1.6)	2.0 (1.4)	< 0.0001
BMI, mean (SD)	29.0 (6.7)	29.0 (6.8)	0.9746	29.8 (7.1)	29.2 (5.9)	0.2646
ARMC, mean (SD)	33.4 (5.0)	32.3 (5.2)	< 0.0001	33.6 (5.2)	32.2 (4.5)	0.0008
WAIST, mean (SD)	98.9 (16.3)	103.3 (16.8)	< 0.0001	101.1 (17.0)	104.8 (14.2)	0.0073
DBP, mean (SD)	71.1 (12.4)	67.4 (15.0)	< 0.0001	70.4 (12.5)	66.7 (16.4)	0.0104
SBP, mean (SD)	122.51 (17.3)	133.7 (22.9)	< 0.0001	125.0 (17.8)	139.5 (23.1)	< 0.0001
Education level, (n%)						
- Less than 9th grade	649 (6.6)	120 (11.5)	< 0.0001	262 (8.0)	14 (10.8)	0.0027
- 9–11th grade	1335 (13.5)	205 (19.7)		343 (10.4)	20 (15.4)	
- High school graduate/GED or equivalent	2158 (21.8)	269 (25.8)		749 (22.8)	37 (28.5)	
- Some college or AA degree	3144 (31.7)	271 (26.0)		1039 (31.6)	43 (33.1)	
- College graduate or above	2625 (26.5)	178 (17.1)		891 (27.1)	16 (12.3)	
Alcohol, mean (SD)	10.6 (24.4)	9.2 (26.5)	< 0.0001	10.0 (22.8)	9.5 (21.9)	0.3346
Fiber, mean (SD)	17.3 (9.4)	15.3(7.9)	< 0.0001	17.0 (9.3)	16.2 (9.6)	0.3440
Energy, mean (SD)	2137.1 (863.0)	1908.8 (768.1)	< 0.0001	2087.2 (828.4)	2124.7 (855.2)	0.6247
Protein, mean (SD)	84.2 (37.4)	74.9 (33.4)	< 0.0001	82.2 (36.1)	79.3 (36.4)	0.3794
Sugar, mean (SD)	113.4 (67.5)	103.0 (61.9)	< 0.0001	102.6 (60.8)	108.7 (58.6)	0.2514
Total fat, mean (SD)	80.1 (39.6)	72.1 (35.0)	< 0.0001	82.3 (40.0)	87.1 (42.2)	0.2047
Cholesterol, mean (SD)	295.6 (194.9)	281.2 (189.5)	0.0196	310.3 (206.4)	328.7 (216.9)	0.3422
Albumin, mean (SD)	28.2 (166.9)	114.6 (462.4)	< 0.0001	44.6 (307.8)	247.9 (818.7)	0.0055
Creatinine, mean (SD)	126.3 (75.5)	108.7 (67.1)	< 0.0001	126.4 (81.8)	123.0 (77.6)	0.6198
Age, mean (SD)	46.4 (16.5)	67.2 (13.9)	< 0.0001	47.9 (16.9)	69.6 (11.0)	< 0.0001

INDHHIN2, Annual household income; INDFMPIR, Ratio of family income to poverty; BMI, Body mass index; ARMC, arm circumference; DBP, diabolic blood pressure; SBP, systolic blood pressure; n (%), frequency (percent).

Number (percent) of death among those with observed feature variables and those with missing study variables from NHANES (2009–2016)-NDI linked mortality (through December 31, 2019) data

		Number (percent) decedents		
Study variables	Total (%) missing	With study variable observed	With study variable missing	P-value
Sex	0 (0)	2232 (9.1)	0 (0)	
Race/ethnicity	0 (0)	2232 (9.1)	(0)	
Marital status	1245 (5.1)	2222 (9.09)	10 (0.04)	< 0.0001
INDHHIN2	1402 (5.7)	2104 (8.6)	128 (0.5)	0.9943
INDFMPIR	2328 (9.5)	2009 (8.2)	223 (0.9)	0.4345
BMI	1163 (4.8)	1987 (8.1)	245 (1.0)	< 0.0001

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(*continued*)

		Number (percent) decedents		
ARMC	2019 (8.3)	1791 (7.3)	441 (1.8)	< 0.0001
WAIST	2265 (9.3)	1735 (7.1)	497 (2.0)	< 0.0001
DBP	1830 (7.5)	1978 (8.1)	254 (1.0)	< 0.0001
SBP	1830 (7.5)	1978 (8.1)	254 (1.0)	< 0.0001
Education level	51 (0.21)	2222 (9.1)	10 (0.04)	0.0094
Alcohol	2724 (11.2)	1861 (7.6)	371 (1.5)	< 0.0001
Fiber	2724 (11.2)	1861 (7.6)	371 (1.5)	< 0.0001
Energy	2724 (11.2)	1861 (7.6)	371 (1.5)	< 0.0001
Protein	2724 (11.2)	1861 (7.6)	371 (1.5)	< 0.0001
Sugar	2724 (11.2)	1861 (7.6)	371 (1.5)	< 0.0001
Total fat	2724 (11.2)	1861 (7.6)	371 (1.5)	< 0.0001
Cholesterol	2724 (11.2)	1861 (7.6)	371 (1.5)	< 0.0001
Albumin	1366 (5.6)	1934 (7.9)	298 (1.2)	< 0.0001
Creatinine	1366 (5.6)	1934 (7.9)	298 (1.2)	< 0.0001
Age	0 (0)	2232 (9.1)	0 (0)	

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